

Kinarm DDM parameter justification

June 23, 2026

Contents

Kinarm DDM parameter justification	2
Bottom line	2
What the three focal parameters mean here	2
Why hierarchical Bayesian estimation fits this problem	3
What can be justified directly from the DDM literature	3
What the reaching and eye-hand literature adds	4
Reaching has a real preparatory floor, but observed reaction time includes extra slack	4
Coordinated eye-hand responses often share a common decision stage but differ after commitment	4
Interception raises urgency and encourages overlap between deciding and acting	4
Nondecision time is not a clean leftover bin	4
Recommended starting bounds and priors for the Kinarm task	5
If the implementation uses $s = 1$	5
If the implementation uses $s = 0.1$	6
Recommended modeling stance by condition	6
Practical implications for the methods section	6
Recommended wording for the core justification	6
Limits of the current justification	7
References	8

Kinarm DDM parameter justification

Bottom line

A diffusion-model analysis of the Kinarm task is defensible if the modeled event is kept narrow: the model should target the **initial commitment to act** rather than the full reach trajectory or later online corrections (Barany et al., 2020; Haith et al., 2016; Ratcliff & McKoon, 2008). For this task, the strongest literature support is not for universal human limits on all DDM parameters, but for a more asymmetric claim: **nondecision time can be partly constrained by physiology and task timing**, whereas **drift rate and boundary separation are more task- and scaling-dependent model parameters** (Bompas et al., 2024; Ratcliff et al., 2016; Tran et al., 2020).

A standard hierarchical DDM is therefore a reasonable starting point for stationary-target trials and a cautious baseline for moving-target interception trials (Vandekerckhove et al., 2011; Wiecki et al., 2013). The broader interception literature suggests that moving targets introduce urgency, overlapping decision and action, and online redirection, so fitted parameters from those trials should be interpreted as summaries of a more complex control process rather than literal isolates of one neural stage (Barany et al., 2020; Bompas et al., 2024; Selen et al., 2012).

What the three focal parameters mean here

Parameter	Working meaning in this project	What the literature supports
Drift rate v	Efficiency and direction of evidence accumulation toward an initial action choice	A valid latent summary of evidence quality in standard DDMs, but its absolute magnitude depends on scaling and task design (Ratcliff et al., 2016; Ratcliff & McKoon, 2008; Tran et al., 2020)
Boundary separation a	Response caution or commitment criterion before movement initiation	A good handle on the speed–accuracy tradeoff, but not a physiological latency bound (Ratcliff et al., 2016; Ratcliff & McKoon, 2008)
Nondecision time T_{er}	Time outside the modeled accumulation process, including encoding, motor preparation/output, and sometimes other unmodeled delays	Useful but not literal; in visuomotor tasks it can absorb several overlapping stages and model mismatch (Bompas et al., 2024; Weindel et al., 2020)

The main practical consequence is that the report should not claim to have found a single set of hard biological limits for all three parameters. The literature supports **outer plausibility bounds** for v and a , and **tighter task-facing timing constraints** for T_{er} (Bompas et al., 2017; Gopal et al., 2015; Haith et al., 2016; Tran et al., 2020).

Why hierarchical Bayesian estimation fits this problem

The hierarchical diffusion framework is well suited to trial-rich but heterogeneous human reaction-time data because it estimates participant-level and group-level parameters together, with shrinkage stabilizing weakly informed individual estimates (Vandekerckhove et al., 2011). This is especially useful when participants differ in speed, caution, and motor timing, or when some conditions have fewer usable trials. The standard HDDM-style approach also gives a natural way to encode prior knowledge about plausible parameter values rather than letting poorly identified parameters drift into implausible regions (Wiecki et al., 2013). That matters most for nondecision time and for any model that includes across-trial variability terms, which are hard to estimate cleanly without strong data support (Böhm et al., 2018; Wiecki et al., 2013).

What can be justified directly from the DDM literature

The best general source for plausible parameter values is the systematic parameter review in (Tran et al., 2020). That review aggregated published DDM fits, rescaled accumulation parameters to the common convention $s = 1$, and proposed informative univariate priors.

Parameter	Suggested prior center or form in the review	Empirical envelope in the review	Use in this project
Drift rate v	Truncated normal dominant component with location about 1.76 and scale about 1.51	0.01 to 18.51	Use as an outer envelope only; absolute values depend on scaling (Tran et al., 2020)
Boundary separation a	Gamma dominant component with shape about 11.69 and scale about 0.12	0.11 to 7.47	Useful as a plausibility check and prior source (Tran et al., 2020)
Nondecision time T_{er}	Truncated t with location about 0.44 s and scale about 0.08 s	0 to 3.69 s	Too broad to use directly for Kinarm reaching without task-specific tightening (Tran et al., 2020)

Two cautions from that review matter immediately.

- All accumulation parameters in that table assume **within-trial noise fixed at $s = 1$** . If a package or custom implementation uses $s = 0.1$, then comparable values for v and a are smaller by a factor of 10 (Ratcliff et al., 2016; Tran et al., 2020).
- The review pools very different tasks. Its priors are therefore useful as **global starting points**, not as task-specific truths (Tran et al., 2020).

A second general caution comes from the standard DDM review: the model is best suited to relatively fast, single-stage decisions, and it treats reaction time as the sum of decision time and nondecision time (Ratcliff & McKoon, 2008). That decomposition is useful here only if the modeled response is the **initial commitment** and not the entire continuous control sequence.

What the reaching and eye-hand literature adds

The Kinarm task differs from canonical button-press DDM tasks because movement unfolds in time and can be adjusted online. The most relevant papers suggest four consistent points.

Reaching has a real preparatory floor, but observed reaction time includes extra slack

In a visually guided reaching task, mean free reaction time was about 212 ms, but movement preparation could be inferred at about 130 ms, leaving an additional initiation delay of about 78 ms on average (Haith et al., 2016). This implies that manual reaction time contains a real physiological floor plus a policy-like initiation margin.

That result supports two claims. First, fitted hand T_{er} values below roughly 130 ms would be hard to defend for this task. Second, a substantial part of measured response time may reflect an initiation policy rather than pure evidence accumulation (Haith et al., 2016).

Coordinated eye-hand responses often share a common decision stage but differ after commitment

A common-accumulator account with effector-specific delay explained coordinated eye-hand reaction times better than interacting separate accumulators in one key study, with hand-specific delay around 160 ms after the shared command and similar eye and hand RT variability in the coordinated condition (Gopal et al., 2015). A follow-up study showed that the architecture is task-dependent: visual-search style coordinated decisions favored a common accumulator, whereas decoupled dual-task settings looked more like separate accumulators (Jana et al., 2017).

For the present project, that means a shared latent decision variable with separate eye and hand output terms is often more defensible than forcing both effectors into one single nondecision parameter (Gopal et al., 2015; Jana et al., 2017).

Interception raises urgency and encourages overlap between deciding and acting

In a robotic reaching and interception task close in spirit to the present setup, interception produced more decision errors, smaller RT increases under added perceptual demands, and more online redirections than stationary reaching (Barany et al., 2020). The interpretation is straightforward: moving targets push the system toward earlier launch and greater overlap between evidence accumulation and movement execution.

This matters because a plain fixed-bound DDM may fit such data numerically while giving parameters that partly stand in for urgency, overlap, or online correction dynamics (Barany et al., 2020; Bompas et al., 2024). That does not rule out DDM use, but it argues for condition-specific interpretation and some restraint in claiming physiological purity for the fitted parameters.

Nondecision time is not a clean leftover bin

The strongest recent warning comes from (Bompas et al., 2024). That paper argues that fitted nondecision time often acts as a mathematical sink for whatever the chosen decision model fails to capture. The authors distinguish biologically grounded visuomotor dead time from model-estimated T_{er} , and they show that the two can diverge strongly.

Single-trial EMG work points the same way. Motor time is not fixed, and it changes with speed–accuracy demands, stimulus strength, and correctness, which means that processes typically assigned to T_{er} are not insulated from decisional manipulations (Weindel et al., 2020). For manual responses, standard assumptions that T_{er} is constant or uniformly variable are therefore simplifications rather than literal truths (Bompas et al., 2024; Weindel et al., 2020).

Recommended starting bounds and priors for the Kinarm task

The table below gives **methods-facing starting values** rather than final truths. These values are appropriate as initial bounds, plausibility checks, or weakly informative priors to be revised after posterior predictive checks.

If the implementation uses $s = 1$

Parameter	Recommended starting range	Confidence level	Justification
Drift rate v	Allow a broad signed range, roughly -18 to $+18$, with most prior mass much closer to 0	Moderate for outer bound, low for task-specific center	Matches the global empirical envelope after allowing both response directions; absolute scale is not portable across implementations (Ratcliff et al., 2016; Tran et al., 2020)
Boundary separation a	Roughly 0.2 to 3 as a practical starting region; hard plausibility ceiling near 7.5	Moderate	The broader empirical envelope extends to 7.47, but most healthy-task fits are much tighter than the full tail (Tran et al., 2020)
Nondecision time T_{er} for hand-only initial commitment	Roughly 0.15 to 0.35 s, with a softer center around 0.20 to 0.28 s	Moderate	Lower end is constrained by reach preparation results and eye-hand delay findings; upper end stays within the range typical of rapid manual visuomotor tasks (Bompas et al., 2017; Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016)
Nondecision time T_{er} for saccade-only initial commitment	Roughly 0.07 to 0.15 s	Moderate	Aligns with saccadic visuomotor dead-time estimates and manual-vs-saccadic comparisons (Bompas et al., 2017; Bompas et al., 2024)

If the implementation uses $s = 0.1$

For v and a , divide the $s = 1$ values above by 10. Time parameters are unchanged (Ratcliff et al., 2016; Tran et al., 2020).

Recommended modeling stance by condition

Condition	Best starting model stance	Why
Stationary-target reaches	Standard hierarchical DDM is defensible	The task is closer to a discrete commitment with less urgency and less need for continuous correction (Ratcliff & McKoon, 2008; Wiecki et al., 2013)
Moving-target interception	Use hierarchical DDM as a baseline, but interpret with caution	Urgency and online redirection can make fixed-bound parameters partly absorb dynamics outside the standard model (Barany et al., 2020; Bompas et al., 2024; Selen et al., 2012)
Joint eye-hand analysis	Prefer a shared decision stage with effector-specific output terms when possible	Coordinated eye-hand tasks often show a common accumulator plus effector-specific delays or noise (Gopal et al., 2015; Jana et al., 2017)

Practical implications for the methods section

A defensible methods section can make the following claims.

- The fitted DDM is intended to capture the **initial decision to launch action**, not the full set of within-movement corrections (Barany et al., 2020; Ratcliff & McKoon, 2008).
- Hierarchical Bayesian estimation is used because it stabilizes participant-level estimates and supports principled prior specification (Vandekerckhove et al., 2011; Wiecki et al., 2013).
- Priors on v and a are anchored to published DDM parameter reviews, with explicit attention to the model scaling convention (Tran et al., 2020).
- Priors or bounds on T_{er} are tightened using manual reaching, saccadic, and eye-hand coordination evidence rather than using a generic global DDM prior alone (Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016; Jana et al., 2017).
- For moving-target trials, parameter interpretation is cautious because urgency and online correction can violate the literal assumptions of a fixed-bound diffusion process (Barany et al., 2020; Bompas et al., 2024; Weindel et al., 2020).

Recommended wording for the core justification

The most defensible short-form justification is this:

A hierarchical drift-diffusion model was used to characterize initial action commitment in the Kinarm task. Drift rate and boundary separation were treated as latent task and policy parameters, with weakly informative priors anchored to published DDM parameter reviews under an explicit scaling convention

(Ratcliff et al., 2016; Tran et al., 2020). Nondecision time was constrained more tightly using visuomotor evidence from reaching, saccadic, and coordinated eye-hand studies, because in such tasks the nondecision component is partly bounded by known sensory and motor delays yet can also absorb model mismatch if left unconstrained (Bompas et al., 2024; Gopal et al., 2015; Haith et al., 2016; Jana et al., 2017; Weindel et al., 2020). Moving-target interception trials were interpreted more cautiously than stationary-target trials because urgency and online redirection can produce overlap between deciding and acting that is only imperfectly captured by a standard fixed-bound diffusion model (Barany et al., 2020).

Limits of the current justification

The literature supports a strong justification for **starting bounds** and for a **cautious interpretation strategy**. It does not support the stronger claim that there are single universal human limits for v , a , and T_{er} that transfer unchanged across tasks, parameterizations, and effectors (Bompas et al., 2024; Ratcliff et al., 2016; Tran et al., 2020).

For this reason, the most defensible workflow is to treat the values above as informed starting constraints, then test them with posterior predictive checks, condition-wise comparisons, and sensitivity analyses on the nondecision prior.

References

- Barany, D. A., Gómez-Granados, A., Schroyer, M., Cutts, S. A., & Singh, T. (2020). Perceptual decisions about object shape bias visuomotor coordination during rapid interception movements. *Journal of Neurophysiology*. <https://doi.org/10.1152/jn.00098.2020>
- Böhm, U., Annis, J., Frank, M., Hawkins, G. E., Heathcote, A., Kellen, D., Kryptos, A.-M., Lerche, V., Logan, G., Palmeri, T., Ravenzwaaij, D. van, Servant, M., Singmann, H., Starns, J., Voss, A., Wiecki, T. V., Matzke, D., & Wagenmakers, E. (2018). Estimating across-trial variability parameters of the Diffusion Decision Model: Expert advice and recommendations. In *Journal of Mathematical Psychology*. <https://doi.org/10.1016/J.JMP.2018.09.004>
- Bompas, A., Hedge, C., & Sumner, P. (2017). Speeded saccadic and manual visuo-motor decisions: Distinct processes but same principles. *Cognitive Psychology*, 94, 26–52. <https://doi.org/10.1016/j.cogpsych.2017.02.002>
- Bompas, A., Sumner, P., & Hedge, C. (2024). Non-decision time: The Higgs Boson of decision. *Psychological Review*. <https://doi.org/10.1037/rev0000487>
- Gopal, A., Viswanathan, P., & Murthy, A. (2015). A common stochastic accumulator with effector-dependent noise can explain eye-hand coordination. *Journal of Neurophysiology*, 113 7, 2033–2048. <https://doi.org/10.1152/jn.00802.2014>
- Haith, A., Pakpoor, J., & Krakauer, J. (2016). Independence of Movement Preparation and Movement Initiation. *The Journal of Neuroscience*, 36, 3007–3015. <https://doi.org/10.1523/JNEUROSCI.3245-15.2016>
- Jana, S., Gopal, A., & Murthy, A. (2017). Evidence of common and separate eye and hand accumulators underlying flexible eye-hand coordination. *Journal of Neurophysiology*, 117 1, 348–364. <https://doi.org/10.1152/jn.00688.2016>
- Ratcliff, R., & McKoon, G. (2008). The Diffusion Decision Model: Theory and Data for Two-Choice Decision Tasks. *Neural Computation*, 20, 873–922. <https://doi.org/10.1162/neco.2008.12-06-420>
- Ratcliff, R., Smith, P. L., Brown, S. D., & McKoon, G. (2016). Diffusion Decision Model: Current Issues and History. *Trends in Cognitive Sciences*, 20, 260–281. <https://doi.org/10.1016/j.tics.2016.01.007>
- Selen, L., Shadlen, M., & Wolpert, D. (2012). Deliberation in the motor system: reflex gains track evolving evidence leading to a decision. *The Journal of Neuroscience : The Official Journal of the Society for Neuroscience*, 32, 2276–2286. <https://doi.org/10.1523/JNEUROSCI.5273-11.2012>
- Tran, N., Maanen, L. van, Heathcote, A., & Matzke, D. (2020). Systematic Parameter Reviews in Cognitive Modeling: Towards a Robust and Cumulative Characterization of Psychological Processes in the Diffusion Decision Model. *Frontiers in Psychology*, 11. <https://doi.org/10.3389/fpsyg.2020.608287>
- Vandekerckhove, J., Tuerlinckx, F., & Lee, M. (2011). Hierarchical diffusion models for two-choice response times. *Psychological Methods*, 16 1, 44–62. <https://doi.org/10.1037/a0021765>
- Weindel, G., Anders, R., Alario, F., & Burle, B. (2020). Assessing model-based inferences in decision making with single-trial response time decomposition. *Journal of Experimental Psychology. General*. <https://doi.org/10.31234/osf.io/kc26f>
- Wiecki, T. V., Sofer, I., & Frank, M. (2013). HDDM: Hierarchical Bayesian estimation of the Drift-Diffusion Model in Python. *Frontiers in Neuroinformatics*, 7. <https://doi.org/10.3389/fninf.2013.00014>